SENIOR PROJECT PROPOSAL · SP1 · 2025



AI Calorie Tracker

AI-Powered Food Recognition & Personalised Nutrition Calculator

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React Native · Expo SDK 54 · Hugging Face API · USDA FoodData Central

taian-an.github.io/calorie-app

github.com/Taian-an/calorie-app

PROBLEM STATEMENT

The Problem

> *"Every time I eat at the cafeteria, I have no idea how many calories I just consumed. Apps like MyFitnessPal require me to manually search every dish — I wanted something simpler:* ***just take a photo.****"*

1. **Manual entry is slow & inaccurate**
   Searching local dish names in English databases returns wrong or missing results.

2. **Barcode scanners don't help**
   Cooked cafeteria meals and homemade food have no barcode to scan.

3. **Generic goals ignore your body**
   A flat "2,000 kcal/day" doesn't account for age, height, weight or activity level.

4. **No combined free solution**
   No free app offers AI photo recognition + personalised TDEE + macro planning together.

PROPOSED SOLUTION

How the App Solves It



AI Food Recognition

Photo → Hugging Face *nateraw/food* model → USDA FoodData Central → calories & macros in under 3 seconds.



Personalised BMR / TDEE

Enter age, gender, height, weight, activity level → Mifflin-St Jeor equation gives your exact daily calorie target.



Macro Planner

Muscle Gain · Fat Loss · Custom — split your TDEE into protein, carbohydrate and fat gram targets with progress bars.

BMR / TDEE Flow

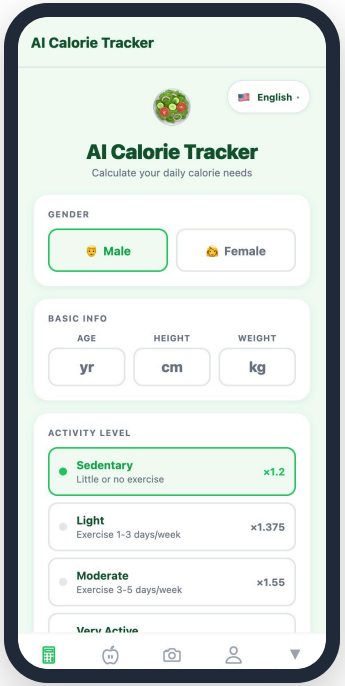
User Input → BMR (Mifflin-St Jeor) → TDEE & Macros

Food Recognition Flow

Photo → Hugging Face (nateraw/food) → USDA Nutrition

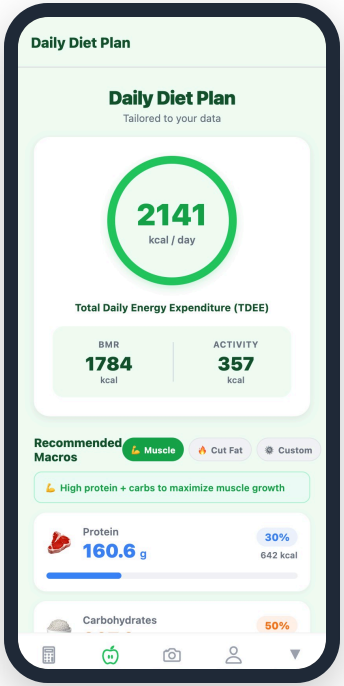
UI / UX DESIGN

App Screens — Live Screenshots



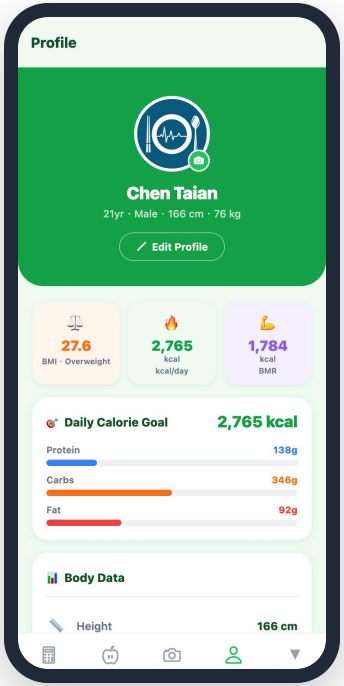
① BMR / TDEE Calculator

Gender · Age · Height · Weight · Activity → calorie goal



② Daily Macro Planner

TDEE ring · BMR · Macro bars · Goal mode selector



③ Profile & Settings

BMI · BMR · daily goal · body data · settings

7 Languages · AI Camera · 3 Goal Modes · Instant Calculation · Macro Breakdown · Profile Page

USE CASES

Two Core User Scenarios

01

AI Food Logging

Camera → Recognition → Nutrition

A student at the cafeteria wants to know the nutrition of their meal without typing anything. They open the camera, point at the food, and tap shutter. The app identifies it via AI and instantly shows calories, protein, fat, and carbs.

expo-camera | Hugging Face API | USDA FoodData

02

Personalised Diet Planning

BMR → TDEE → Macro Targets

A gym-goer starting a muscle-gain programme needs their daily calorie and macro targets. They enter body data, select "Muscle Gain" mode, and receive protein, carb, and fat targets calculated from their actual TDEE.

Mifflin-St Jeor | TDEE Calculation | Macro Planner

USE CASE 01

AI Food Logging — Step by Step

Actor: Student at cafeteria · **Goal:** Get nutrition without typing

1. **Open Camera Tab**
   Navigate to 📷 Camera. App requests camera permission on first launch.

2. **Frame Food & Tap Shutter**
   Point phone at meal → tap circular shutter button. Photo captured locally.

3. **AI Recognition**
   Photo sent to *nateraw/food* via Hugging Face API → returns label + confidence, e.g. `"fried_rice — 94%"`.

4. **Nutrition Lookup**
   Food label queried against USDA FoodData Central → Energy, Protein, Carbs, Fat per 100g.

5. **Result Card Shown**
   Result card slides up with food name, confidence badge, 4 macro tiles. Total: ~2–3 seconds.

USE CASE 02

Personalised Diet Planning — Step by Step

Actor: Gym-goer (muscle-gain) · **Goal:** Get personalised macro targets

1. **Open Calculator Tab**
   Navigate to Calculator tab → select gender (Male / Female).

2. **Enter Body Data**
   Type age, height (cm), weight (kg). All three fields required before calculation.

3. **Select Activity Level**
   Choose from 5 levels: Sedentary ×1.2 → Lightly Active ×1.375 → Moderate ×1.55 → Very Active ×1.725 → Athlete ×1.9.

4. **View TDEE Result**
   Mifflin-St Jeor: BMR = 88.36 + 13.4×kg + 4.8×cm – 5.7×age. TDEE = BMR × factor. Shown in ring gauge.

5. **Choose Goal Mode**
   Tap **Muscle Gain** → Protein 30% · Carbs 50% · Fat 20% — gram amounts & kcal shown with progress bars.

Demo / Screenshot

RECAP

Summary

🎯 Problem

Manual food logging is slow & inaccurate. Generic calorie goals ignore individual body data. No free app combines both.

💡 Solution

AI photo recognition (Hugging Face × USDA) + science-based personalised TDEE (Mifflin-St Jeor) in one cross-platform app.

✅ Technical Feasibility

Both APIs integrated & tested. Web build live at **taian-an.github.io/calorie-app**. Public GitHub repo.

📱 UI/UX Design

4 screens: Calculator · Planner · Camera · Profile. Mobile-first, green design system, 7-language support.

Addressing Reviewer Feedback

Previously: Technical feasibility not demonstrated
Now: Live demo + real API code + public GitHub repo

Previously: No UI/UX design shown
Now: Screenshots of all screens from the running build

Previously: Slides mostly AI-generated
Now: Personal motivation, real errors & fixes, specific technical decisions

🌐 **taian-an.github.io/calorie-app**

github.com/Taian-an/calorie-app